

Customer Churn Analysis Report

SQL + Python Analysis of Subscription Customer Data
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Executive Summary

This report summarizes a churn analysis of a subscription-based customer base, built by integrating customer, subscription, and support data from a relational database. The analysis quantifies overall churn, breaks it down by plan type and acquisition channel, and identifies support escalations as the strongest measurable driver of churn. The dataset covers 21 customers and should be read as a directional analysis rather than a statistically exhaustive study; the patterns below are nonetheless clear and actionable.

Key Performance Indicators (KPIs)

Metric	Value
Overall Churn Rate	28.57%
Retention Rate	71.43%
Average Revenue Per User (ARPU)	\$18.85 / month
Monthly Revenue at Risk (churned customers)	\$73.94
Support Escalation Rate	19.05%
Correlation: Escalations vs. Churn	76.7%
Churn Rate — Basic Plan	60.00%
Churn Rate — Standard Plan	22.22%
Churn Rate — Premium Plan	14.29%
Churn Rate — Referral-Acquired Customers	83.33%
Churn Rate — Paid-Acquired Customers	16.67%
Churn Rate — Organic-Acquired Customers	0.00%

Insights

1. Escalations Are the Strongest Churn Signal

Customers with an escalated support case churn at a substantially higher rate than those without one, with a 76.7% correlation between escalation status and churn — by far the strongest relationship found in the data.

2. Churn Falls Sharply as Plan Tier Rises

Basic plan customers churn at 60%, more than double the Standard plan rate (22.22%) and over four times the Premium rate (14.29%). Higher-tier plans appear to retain customers meaningfully better.

Plan Type	Customers	Churn Rate	Total Revenue
Basic	5	60.00%	\$52.95
Standard	9	22.22%	\$123.91
Premium	7	14.29%	\$218.93

3. Acquisition Channel Strongly Predicts Retention Quality

Referral-acquired customers churn at 83.33%, compared to 16.67% for Paid and 0% for Organic customers. This is the widest gap observed across any segment in the analysis.

Acquisition Channel	Customers	Churn Rate	Total Revenue
Referral	6	83.33%	\$74.94
Paid	6	16.67%	\$174.94
Organic	9	0.00%	\$145.91

4. Revenue Exposure Is Currently Modest but Concentrated

Average revenue per user is \$18.85/month, with \$73.94 in monthly recurring revenue already lost to churned customers. While the absolute figures reflect the current dataset's small scale, the proportional exposure (roughly 29% of the customer base) mirrors the overall churn rate and would scale directly with a larger customer base.

Action Items

Priority	Action Item	Rationale	Owner
High	Review the Referral acquisition/onboarding experience	Referral customers churn at 83.33% — the highest of any channel	Marketing / Onboarding
High	Trigger proactive retention outreach on any escalated support case	76.7% correlation between escalation and churn is the strongest signal in the data	Customer Success
Medium	Target retention offers at Basic plan subscribers	Basic plan churns at 60%, vs. 14.29% for Premium	Retention / CRM
Medium	Monitor churn by plan tier and acquisition channel monthly	These two dimensions show the clearest separation in the current dataset	Analytics

Low	Validate findings against a larger customer sample	Current analysis is based on 21 customers — directional, not yet statistically robust	Analytics
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Limitations & Next Steps

This analysis is based on a small demonstration dataset (21 customers) intended to validate a full SQL-to-Python churn analysis workflow. Segment-level findings (e.g., state-level churn) should be treated as directional given small sub-group sizes. Recommended next steps: replicate segment aggregations as native SQL queries, incorporate Customer Lifetime Value (CLTV) data already available but unused, build a cohort/retention-curve view, and validate findings against a larger, real-world dataset before treating them as final.